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Problem Set 4

1.

a. calculate the first 10 triangles

T(10) = T(0) + T(1) + T(2) + T(3) +T(4) + T(5) + T(6) + T(7) + T(8) + T(9)

T(0) = 1

T(1) = 1 + 2 = 3

T(2) = 1 + 2 + 3 = 6

T(3) = 1 + 2 + 3 + 4 = 10

T(4) = 1 + 2 + 3 + 4 + 5 = 15

T(5) = 1 + 2 + 3 + 4 + 5 + 6= 21

T(6) = 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28

T(7) = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36

T(8) = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45

T(9) = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55

sum of triangles starting at i = 0 : = $\sum_{i=0}^{9}(n+1)$

sum of triangles starting at i = 1 : = $\sum_{i=1}^{10}(n+1)$

b. Show the first 5 elements of each series. In the table below, the series for each summation is started at either n= 0 or n = 1. In the case where n = 0, the last element in the series will be n= 4 and would include the numbers 0, 1, 2, 3, 4 for a total of 5 elements. When n is incremented from 1, the ending value will be 5. Note that the expression changes for each case to compensate for the difference in starting values.

	$\boldsymbol{n = 0}$	$\boldsymbol{n = 1}$
Sums of Integers	$\sum_{n=0}^{n=4}(n+1)$ $n(0) = 1$ $n(1) = 2$ $n(2) = 3$ $n(3) = 4$ $n(4) = 5$ $\therefore \sum_{n=0}^{n=4}(n+1) = 5+4+3+2+1 = 15$	$\sum_{n=1}^{n=5}(n)$ $n(1) = 1$ $n(2) = 2$ $n(3) = 3$ $n(4) = 4$ $n(5) = 5$ $\therefore \sum_{n=1}^{n=5}(n) = 5+4+3+2+1 = 15$
Sums of Odds	$\sum_{n=0}^{4}(2n+1)$ $n(0) = 1$ $n(1) = 3$ $n(2) = 5$ $n(3) = 7$ $n(4) = 9$ $\therefore \sum_{n=0}^{4}(2n+1) = 1+3+5+7+9 = 25$	$\sum_{n=1}^{n=5}(2n-1)$ $n(1) = 1$ $n(2) = 3$ $n(3) = 5$ $n(4) = 7$ $n(5) = 9$ $\therefore \sum_{n=1}^{n=5}(2n-1) = 1+3+5+7+9 = 25$
Sums of Cubes	$\sum_{n=0}^{n=4}(n+1)^3$ $n(0) = 1^3 = 1$ $n(1) = 2^3 = 8$ $n(2) = 3^3 = 27$ $n(3) = 4^3 = 64$ $n(4) = 5^3 = 125$ $\therefore \sum_{n=0}^{n=4}(n+1)^3 = 1+8+27+64+125 = 225$	$\sum_{n=1}^{n=5}(n)^3$ $n(1) = 1^3 = 1$ $n(2) = 2^3 = 8$ $n(3) = 3^3 = 27$ $n(4) = 4^3 = 64$ $n(5) = 5^3 = 125$ $\therefore \sum_{n=1}^{n=5}(n)^3 = 1+8+27+64+125 = 225$

c. Closed form of the sum of integers. The following table shows the results of the sum and closed form of the first 5 elements of the sum of integers. Note that if the bounds included 0 (instead of starting at 1), n(0) would result in 0 which would not alter the results of any subsequent summations therefore the formula would not have to change.

	$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$n = 1$	$\sum_{k=1}^{n} 1 = 1$	$\frac{1(1+1)}{2} = 1$
$n = 2$	$\sum_{k=1}^{n} 1 + 2 = 3$	$\frac{2(2+1)}{2} = 3$
$n = 3$	$\sum_{k=1}^{n} 1 + 2 + 3 = 6$	$\frac{3(3+1)}{2} = 6$
$n = 4$	$\sum_{k=1}^{n} 1 + 2 + 3 + 4 = 10$	$\frac{4(4+1)}{2} = 10$
$n = 5$	$\sum_{k=1}^{n} 1 + 2 + 3 + 4 + 5 = 15$	$\frac{5(5+1)}{2} = 15$

2. Find the sets A and B if A – B = { 1, 5, 7, 8}, B - A = {2, 10}, and A∩B = {3, 6, 9}. The definition of difference of A and B (denoted as "A-B") is that the elements of A are not in the set of B. Therefore, the values 1, 5, 7, 8 cannot be contained in B. Similarly, the values 2, 10 cannot be in set A. The ending statement A∩B can be read as A intersection of B, or that the values of A are also found in B therefore the values 3, 6, 9 are found in both sets A and B. The sets A and B are:

   A = {1,3,5,6,7,8,9}

   B = {2,3,6,9,10}

3. Cardinality

a. $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Step	Action	Justification
1	$\lvert A \cup B \cup C \rvert$	Premise
2	$= \lvert A \cup (B \cup C) \rvert$	Associative law
3	$= \lvert A \rvert + \lvert B \cup C \rvert - \lvert A \cap (B \cup C) \rvert$	Principle of Inclusion Exclusion (PIE) A and $B \cup C$
4	$\lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert$	PIE $B \cup C$
5	$= \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert - \lvert A \cap (B \cup C) \rvert$	Sub step 4 into step 3
6	$= \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert - \lvert A \cap B \mid \cup \lvert (A \cap C) \rvert$	Expand union
7	$\lvert A \cap B \mid + \lvert (A \cap C) \rvert - \lvert A \cap B \mid \cap \lvert (A \cap C) \rvert$	PIE $\lvert A \cap B \mid \cup \lvert (B \cap C) \rvert$
8	$= \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert - (\lvert A \cap B \mid + \lvert (B \cap C) \rvert - \lvert A \cap B \mid \cap \lvert (A \cap C) \rvert)$	Substitute 7 into step 6
9	$= \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert - \lvert A \cap B \mid - \lvert (B \cap C) \rvert + \lvert A \cap B \mid \cap \lvert (A \cap C) \rvert$	Get rid of parenthesis
10	$\lvert A \cap B \cap C \rvert$	Union of $\lvert A \cap B \mid \cap \lvert (B \cap C) \rvert$
11	$= \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert B \cap C \rvert - \lvert A \cap B \mid - \lvert A \cap C \rvert + \lvert A \cap B \cap C \rvert$	Insert Step 10 and Finito!

b. Kiddos that passed the exam

Let C be the number of students that failed Computer Science, M the number of students that failed Applied Mathematics and P be the number of students that failed Philosophy.

Step	Action	Justification
1	$\lvert -C \cap -M \cap -P \rvert$	Premise
2	$= \lvert -(C \cup M \cup P) \rvert$	De Morgans law
	$= U - C \cup M \cup P$	Complement law (let U = universe)
3	$= U - (\lvert C \rvert + \lvert M \rvert + \lvert P \rvert - \lvert M \cap P \rvert - \lvert C \cap M \mid - \lvert C \cap P \rvert + \lvert C \cap M \cap P \rvert)$	Step 11 in part 3.a.
4	$= U - (12 + 5 + 8 - 3 - 2 - 6 + 1)$	Substitute given cardinalities of each set
5	$= 41 - (15) = 16$	16 students did not fail C or M or P (passed all)

4. one-to-one and/or onto

a. $f(n) = n^3 - 1$ where f: Z -> Z

n	F(n)
100	-1000001
-2	-9
0	-1
1	0
2	7
100	999999

For every possible value of n that is an integer, f(n) is different however, not every integer is used therefore this equation is one-to-one but not onto.

b. f(m,n) = 2m – n where f: ZxZ -> Z

m	n	2m - n
-3	3	-9
-2	2	-6
-1	1	-3
0	0	0
1	-1	3
2	-2	6
3	-3	9
4	4	4
5	5	5
6	6	6
-4	-4	-4
-5	-5	-5
-6	-6	-6
1	2	0
2	1	3

For every possible value of integers n, f(n) may be pointed to by different pairs of m,n (highlighted cells above) and not every integer available is used therefore this equation is NEITHER one-to-one or not onto.